

CHAPTER 21

LEAD

- Ref:
- (a) 29 CFR
 - (b) OPNAVINST 5100.19F
 - (c) Industrial Hygiene Field Operations Manual NMCFHPC-TM6290.91-2
 - (d) American Conference of Governmental Industrial Hygienists Pub. No. 2096, Industrial Ventilation: A Manual of Recommended Practice, 29th Edition
 - (e) American National Standards Institute (ANSI) Z9.2-2018, American National Standard for Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems
 - (f) UFC 3-410-04N of 13 December 17, Design: Industrial Ventilation
 - (g) Centers for Disease Control and Prevention, Guidelines for the Identification and Management of Lead Exposure in Pregnant and Lactating Women, November 2010
 - (h) NMCPHC Technical Manual OM 6260, Medical Surveillance Procedures Manual and Medical Matrix, Aug 2015
 - (i) DoD Instruction 6055.05, Occupational and Environmental Health (OEH), 31 August 2018
 - (j) UFGS-02 83-0, Lead in Construction, November 2018

B2101. Discussion.

a. The goal of this chapter is to prevent employee lead exposure and related injuries during the use, handling, removal and melting of materials containing lead at Navy commands, units and activities. The provisions of this chapter and reference (a) sections 1910.1025, 1915.1025 and 1926.62 apply to lead exposures in general industry, shipyards, shipbreaking and construction work.

(1) Construction work covered by reference (a) section 1926.62, includes any repair, renovation or other commands, units and activities that disturb in place lead-containing materials (LCM) (e.g., removal of lead paint in way of steel structure renovation and repair), but does not include routine cleaning and repainting (e.g., minor surface preparation and repainting of rental apartments between tenants or at scheduled intervals) where there is insignificant damage, wear or corrosion of existing lead-containing paint, coatings or substrates.

(2) Employees performing processes not associated with construction work which may have the potential to cause occupational exposure to personnel are covered by the general industry standard for lead, reference (a) section 1910.1025. Maintenance commands, units and activities covered by the general industry standard are those that involve making or keeping a structure, fixture or foundation in proper condition in a routine, scheduled or anticipated fashion.

b. Navy commands, units and activities with ashore operations must establish a lead control program that complies with Federal Occupational Safety and Health Administration (OSHA) regulations (reference (a) sections 1910.1025, 1915.1025 and 1926.62) applicable to the type of operation. This includes operations performed aboard vessels while in port or a repair facility. Reference (b) discusses the lead control program for forces afloat.

B2102. Program Definitions and Health Effects.

a. Lead, as used in this chapter, means metallic lead, all inorganic lead compounds and organic lead soaps (e.g., lead naphthenate). Lead's abundance, low melting point, high molecular weight, high density and malleability make it a useful material. When added to resins, grease or rubber, lead compounds act as antioxidants. Although lead has been banned from residential paint, lead and other toxic metals can be found in industrial and maritime paints, coatings and older housing (pre-1978). Other common uses for lead and lead compounds include:

- (1) Ballast
- (2) Radiation shielding
- (3) Paint filler and hardener
- (4) Rubber antioxidant
- (5) An acoustical insulation component
- (6) Solder for electrical components and pipe joints
- (7) High voltage cable shielding
- (8) Small arms ammunition
- (9) Batteries
- (10) Roof flashing
- (11) Weights

b. Significant lead exposures can occur during:

(1) Melting lead or lead-containing Babbitt (a mixture containing two or more metallic elements or metallic and nonmetallic elements usually fused together or dissolving into each other when molten).

- (2) Casting molten lead or alloy containing lead
 - (3) Ballast handling.
 - (4) Cutting ballast straps with torch or grinder.
 - (5) Spraying, sanding, scraping, grinding, burning, welding and abrasive blasting, especially on components and equipment with lead based coatings or lead alloys.
 - (6) Bullet trap clean-out and general cleaning at firing ranges.
 - (7) Lead shield repair.
 - (8) Repairing electronics with lead solder.
 - (9) Special Hull Treatment (SHT) tile cutting.
 - (10) High voltage cable repair.
 - (11) Lead-acid battery reclaiming.
 - (12) Cleanup or handling of lead-contaminated debris and clothing.
- c. Lead is a recognized health hazard. Lead overexposure can damage many organs and systems, especially the brain, central nervous system, peripheral nervous system, kidneys, reproductive system and inhibit heme synthesis or erythropoiesis (red blood cell production). The Navy recognizes the serious health hazards associated with lead exposure and has established strict controls for processes which can potentially expose personnel to these hazards.
- B2103. Methods to Control Potential Airborne Lead Hazards. Chapter 12 discusses the basic principles for controlling hazards in the occupational environment including substitution with less hazardous materials, engineering controls (closed systems, thermostats), administrative controls (job rotation, work time limits) and use of personal protective equipment (PPE), in that order. Total reliance on PPE is acceptable only when all other methods are proven to be technically or economically infeasible.
- a. General Workplace Control Practices.
- (1) The Navy must not use paints containing more than 0.009 percent lead by dry weight unless the cognizant headquarters command for the command, unit and activity responsible for the work specifically approves higher lead content paint and documents approval criteria.

(2) Before proceeding with work involving paint removal for surface preparation or in the way of hot work, commands, units and activities must determine if lead is a constituent of the paint. This may be accomplished via handheld XRF analyzers, testing of the paint using a valid laboratory method with results in percent lead by weight of paint or through established and accurate records which provide the needed information (e.g., paint application records, prior testing results or other valid documentation). The cognizant safety professional or industrial hygienist (IH) must determine the lead monitoring and controls required for the work. This determination must be based on the lead content of the involved paints, the work methods to be employed and observation, calculations or previous measurements relevant to the employee exposure potential of the work in question.

(3) When feasible, commands, units and activities must minimize the heating of lead and leaded materials through the use of thermostatically controlled heating or the removal of lead containing coatings prior to heating.

(4) Commands, units and activities must establish procedures to maintain work surfaces as free of lead dust as practicable and must clean up lead dust with high efficiency particulate air (HEPA) filtered vacuum cleaners. Compressed air is prohibited for cleaning floors or other surfaces. Wet or dry sweeping, shoveling or brushing will only be used when a HEPA filter vacuuming or other equally effective methods that minimize the likelihood of exposure have been tried and do not work. Industrial hygiene (IH) will consult and approve the alternative cleaning method(s). Refer to Chapter 12, subparagraph B1203j of this Manual for additional discussion and requirements related to establishing and sustaining housekeeping and hygiene practices to maintain all surfaces as free as practicable of surface accumulations at operations generating airborne dust, mist or aerosols containing lead.

(5) Commands, units and activities that have lead containing waste, scrap, debris, containers, equipment and clothing consigned for disposal must collect it, seal it in impermeable containers and label waste per subparagraph B2103e.

(6) To minimize exposure potential, commands, units and activities must isolate hot work on lead and abrasive lead removal operations to the extent feasible, from other operations.

(7) Commands, units and activities whose employees have occupational exposure to lead must have a written compliance program specific to their command, unit or activity which meets the requirements of reference (a) section 1910.1025.

b. Ventilation. Mechanical exhaust ventilation is frequently required to ensure that atmospheric levels of lead particulate do not exceed the Permissible Exposure Limit (PEL). General requirements for the design and use of ventilation systems to reduce exposures are listed in subparagraph B2103b. The cognizant industrial hygienist must provide specific guidance for each lead operation.

(1) Commands, units and activities must design, construct and maintain local exhaust ventilation and dust collection systems using guidelines in reference (c) through (g) or equivalent, with assistance from cognizant IH personnel and facilities engineers.

(2) Commands, units and activities must coordinate with cognizant IH for testing of ventilation systems used to control lead exposures or emissions using qualified engineering or IH personnel at least every 3 months and within 5 days of any production, process or control change which may result in a change in employee exposure. Where devices such as manometers, pitot tubes (a device to measure fluid flow velocity), etc., are installed to continuously monitor the effectiveness of ventilation systems, commands, units and activities must instruct employees who use the system on the meaning and importance of the measurements and to immediately contact their safety office if the measuring devices indicate a malfunction. Where such devices are in place, IH or engineering personnel only need to inspect the ventilation systems annually.

(3) Commands, units and activities using ventilation systems to control occupational exposures or emissions must not re-circulate air from operations generating lead into any workspace. Ventilation is to be HEPA filtered before exhausted directly to the atmosphere. The command, unit or activity environmental manager must approve the air pollution control system after consulting with the cognizant air pollution regulatory agency.

(4) The industrial hygienist must review the ventilation design for ease of maintenance and accessibility, as well as design errors and will pay special attention to hoods, duct work, clean out hatches, exhaust fans and air pollution control devices. Commands, units and activities must install the exhaust fan, after the air pollution control system, in a protected and restricted room or shed. If a HEPA filter is required and the filter and pre-filter housing is located outdoors, they must use a bag-in, bag-out style access housing.

Note: Ventilation design review may not be required for indoor firing range repair or modification under the technical support of the Naval Facilities Engineering Systems Command's (COMNAVFACSYSCOM) Technical Center of Expertise program if using a performance-based contracted approach and the design has been certified by COMNAVFACSYSCOM's Technical Center of Expertise.

c. Personal Protective Clothing and related control facilities.

(1) Personnel engaged in operations where the concentration of airborne lead particulates is likely to be equal to or exceed the PEL or where the possibility of skin or eye irritation exists, must remove clothing worn to and from work and wear the protective clothing provided by the Navy. Employees must use waterproof clothing when handling wet lead. Protective clothing includes:

(a) Full body, one-piece coveralls.

(b) Durable gloves and head covering or other appropriate protective equipment which complies with reference (a) section 1910.1025. Hoods must extend beyond the collar of the coverall, completely protecting the neck area.

(c) Slip-resistant shoe covers or lightweight rubber boots and may also use disposable shoe covers.

(d) Vented safety goggles may be worn in lieu of safety glasses with side shields. When there is the potential for a splash to occur or debris to get on employee's face, a face shield will be made available to be worn over safety glasses or goggles.

Note: The proper use of protective clothing requires that employees close all openings and that garments fit snugly about the neck, wrists and ankles. Accordingly, employees must tape the wrist and ankle junctions, as well as the collar opening on coveralls as necessary, to prevent contamination of skin and underclothing without restricting physical movement.

(2) Commands, units and activities must provide change rooms as close as practical to the lead work area(s) for employees who work where the airborne lead exposure is above the PEL (without regard to the use of respirators). They must maintain change rooms under positive pressure with respect to adjacent lead work areas. They must post protective clothing removal procedures in the change room and include vacuuming of clothing (before removal and while still wearing a respirator, if one was required for the task) using a HEPA filter vacuum. Removal of lead particles from clothing by blowing or shaking is prohibited.

(3) Contracts governing laundering of lead-contaminated clothing must specifically require that contractors comply with the precautions specified in reference (a) section 1910.1025.

d. Limits of Respirator Usage

(1) Commands, units and activities must use engineering control measures per this chapter and will not achieve compliance with PELs solely by the use of respirators except under these conditions:

(a) During the time period necessary to implement engineering control measures.

(b) In work situations in which the control methods prescribed are not technically feasible or are not sufficient to reduce the airborne concentration of lead particulates below the PEL.

(c) During emergencies.

(2) Where respirators are required to control exposure to lead, commands, units and activities must comply with the respirator program per Chapter 15 of this Manual and reference (a) sections 1910.1025, 1915.1025, 1926.62 and reference , (c). Negative pressure air-purifying

respirators must be equipped with P100 filters and powered air purifying respirators (PAPRs) must be equipped with high efficiency particulate air (HEPA) filters.

(3) Commands, units and activities must provide a respirator to employees who work with lead, upon request and must enter the employee into the respiratory protection program.

e. Warning Signs and Labels. Commands, units and activities must affix caution labels to containers of lead contaminated clothing, equipment, raw materials, waste, debris or other products per reference (a) section 1910.1025.

f. Lunch Rooms and Personal Hygiene. Lunch rooms and personal hygiene must be per OSHA rules and regulations.

B2104. Dust Accumulation Requirements. Commands, units and activities must establish and sustain housekeeping and hygiene practices to maintain all surfaces as free as practicable of surface accumulations at operations generating airborne dust, mist or aerosols containing lead, hexavalent chromium, cadmium or beryllium. “As free as practicable” is defined as conducting regular cleaning and housekeeping activities to prevent avoidable dust exposure, such as those caused by potentially re-entrained dust. Commands, units and activities must employ a systems engineering approach to establish, document and integrate refined procedures to ensure surfaces are maintained as free as practicable of accumulations of lead, hexavalent chromium, cadmium or beryllium by:

a. Establish housekeeping plans, schedules and procedures. Compressed air is prohibited for cleaning floors or other surfaces. High efficiency particulate air (HEPA) vacuuming will be the primary method for maintaining surfaces as free as practicable of accumulations. Wet or dry sweeping, shoveling or brushing will only be used when HEPA-filter vacuuming or other equally effective methods that minimize the likelihood of exposure have been tried and do not work. IH must be consulted and approve these alternative cleaning methods.

b. Establish containment processes and methods to monitor their effectiveness.

c. Implement best practices decontamination procedures.

d. Establish engineering controls tailored to each facility.

e. Sustain compliance actions, as appropriate, in the local written operating procedures and quality control procedures.

f. Include all key stakeholders in these process improvement activities to ensure the broadest range of solutions as well as widest acceptance during implementation. SOH personnel have an integral role in the development of these process improvement activities.

B2105. Environmental Protection Procedures.

- a. Commands, units and activities must invoke local, state and federal environmental standards including sections 1910.1025 and 1915.1025 of reference (a), reference (b) and reference (g) or requirements of paragraph B2103 of this chapter, whichever are more stringent.
- b. Commands, units and activities must require salvageable lead and lead waste be contained and labeled as required by local, state and federal environmental requirements. Bagging in heavy duty plastic bags or impermeable containers labeled with caution labels identified in subparagraph B2103e will be the minimum method of containment of lead salvage and waste until otherwise directed by the cognizant environmental manager for the command.
- c. Commands, units and activities cognizant environmental manager must invoke requirements of local, state and federal requirements related to lead emissions.
- d. Technical assistance for air pollution control is available upon request from the COMNAVFACSYSCOM Facilities Systems Engineering Commands.

B2106. Exposure Monitoring. An exposure-monitoring plan must be established for any lead operation with the potential to cause employee exposure at or above the action level. Qualified persons will conduct initial and periodic monitoring. Persons qualified to perform exposure monitoring are specified in Chapter 8 of this Manual. Cognizant IHs will conduct exposure assessments as outlined in Chapter 8 of this Manual. See B0803.a for requirements on documentation of exposure monitoring results in each employee's medical record.

B2107. Medical Surveillance.

- a. General. This program consists of three basic elements:
 - (1) Pre-placement medical evaluation.
 - (2) Semi-annual Blood lead monitoring.
 - (3) Follow-up medical evaluations based on the results of blood lead analysis and physician opinion.
 - (4) Unless a medical evaluation was completed within the past twelve months, a termination examination identical in scope to the baseline examination must be conducted just prior to the reassignment or termination of a person from a job requiring medical surveillance due to lead exposure. Commands, units and activities must include personnel in this program when IH survey report indicates that they perform work or are likely to be in the vicinity of an operation which generates airborne lead concentrations at or above the AL for more than 30 days per year. Examinations may include special purpose histories and physical examinations and

laboratory tests designed to detect early signs of lead over-absorption. Refer also to reference (h) for medical protocols and guidance. Commands, units and activities must base inclusion into this program on airborne concentration measurements without regard to respirator use and, therefore, inclusion does not indicate that an individual is overexposed to lead.

b. Program Elements

(1) Pre-placement Evaluation. All Navy personnel who meet criteria of subparagraph B2106a(4) must receive a pre-placement evaluation as described in reference (h) prior to assignment to a position involving potential exposures to lead that equal or exceed the AL.

(2) Blood Lead Levels and Frequency of Monitoring. Navy commands, units and activities must make blood lead analysis and zinc protoporphyrin (ZPP) available every 6 months for all personnel who are or may be exposed to lead above the AL for more than 30 days per year. Supporting medical facilities must perform analysis per reference (i).

(3) Follow-up Medical Surveillance

(a) Individual Reassignment or Medical Removal. An employee must be reassigned to non-lead work:

1. If an employee's blood lead concentration equals or exceeds the levels specified in reference (i).
2. If the occupational medicine provider recommends removal.
3. Or, if the employee has signs or symptoms of lead toxicity.

Note: Additional guidance concerning removal procedures, return to former job status and removal protection requirements is contained in reference (a) section 1910.1025 and (i). Commands, units and activities must also reassign pregnant women exposed to lead at or above specified levels or who have a blood lead level of 5 ug/100g blood (refer to reference (f)) to a job without lead exposure, with medical removal benefits.

(b) Follow-up Blood Lead Monitoring. Commands, units and activities must perform follow-up lead monitoring as specified in reference (i) and periodically thereafter according to this criteria. During medical removal (to non-lead work activity), commands must monitor the employee's blood lead concentration monthly until the employee's last two consecutive test results are at or below the level specified in reference (i), at which time the employee may be returned to their regular work activity.

(c) Follow-up Evaluations

1. Medical Follow-up. Commands must conduct a medical evaluation per reference (i). The medical department must notify the cognizant IH of each elevated blood lead level which has been verified by follow-up blood lead monitoring.

2. Reassignment - Termination of Employment Follow-up. Commands, units and activities must conduct a medical evaluation identical to the pre-placement evaluation just prior to the reassignment or termination of an employee from a job requiring medical surveillance, unless a medical evaluation was done within the past twelve (12) months.

3. Physicians Written Opinion. Reference (a) section 1910.1025 requires a written opinion and reference (h) provides a sample written opinion.

4. IH Follow-up Investigation. When notified of an elevated blood lead level, perform an investigation to determine whether there is an occupational cause for follow-up blood lead monitoring.

(d) Other Appropriate Medical Evaluations. The cognizant medical activity must perform a medical examination, including those elements of the pre-placement examination, which the physician deems necessary:

1. As soon as possible after notification by an employee that they have developed signs or symptoms commonly associated with lead intoxication.

2. As soon as possible after notification that the employee desires medical advice concerning the effects of current or past lead exposure on the ability to procreate a healthful child.

3. As soon as possible after being informed that the employee demonstrates difficulty breathing during a respirator fit test or during respirator use.

4. As medically appropriate for personnel who have been removed from exposure to lead due to risk of sustaining material impairment to health or otherwise limited pending a final medical evaluation.

5. As a result of airborne exposure monitoring results.

c. Administrative Procedures

(1) Employee Notification. Commands, units and activities must notify the employee in writing, within 5 working days after receipt of results, when the blood lead level is at or above the level specified in reference (i):

(a) The blood lead concentration level, as reported.

(b) That the regulations require temporary medical removal with Medical Removal Protection benefits when and if, the employee's blood lead level exceeds the current numerical criterion for medical removal under reference (a) section 1910.1025 and reference (i), whichever is lower.

(2) Employee Counseling. The physician must counsel personnel regarding any abnormalities detected during any screening test. The physician must make an entry into the employee's medical record that describes the counseling given. The employee must countersign this entry. A copy of the physician's written opinion countersigned by the employee can satisfy this requirement and reference (h) provides a sample written opinion with space for an employee signature.

d. Medical Records

(1) Each employee Occupational Health Medical Record (OHMR) will include this identifying information:

(a) Name

(b) Date of birth

(c) Dates of examinations

(d) Job titles, job codes or primary and secondary Navy Enlisted Classification Code (NEC).

(2) All records of examinations, possible lead-related conditions, related laboratory results, airborne exposure monitoring results and all forms and correspondence (including the physician's written opinions) related to the employee's medical history must become a permanent part of the employee health record. The cognizant medical activity must retain such records for the period of employment plus 20 years or 40 years, whichever is longer.

(3) Medical facilities must enter the judgment of the occupational health physician concerning the adequacy of the diagnostic information to support the impression of lead-related disease in the medical record. Lacking definitive information, the evaluating physician must exercise their best medical judgment on each individual case.

(4) The Commands, units and activities cognizant medical authority must make available copies of any examinations, laboratory results or special studies in an employee's health record or compensation folder to a physician of the employee's choice after execution of a proper release of information form.

(5) Should the Navy select the initial physician, the employee may designate a second physician to review any findings and conduct independent examinations and tests as may be deemed necessary. The Navy must provide to the initial and consulting (second) physician copies of:

- (a) Reference (a) section 1910.1025 and this chapter.
 - (b) Description of employee's duties.
 - (c) Employee's exposure level.
 - (d) Description of PPE.
 - (e) Blood lead determinations.
 - (f) All prior written medical opinions.
- (6) The cognizant medical activity must maintain these medical records.

(7) Each individual currently or previously employed by Department of the Navy or any other person they designate must have access to the records within 15 days of the request.

B2108. Work Performed by Private Contractors.

a. Reference (j) should be used on construction projects impacting material containing lead or paint with lead. Use reference (j) in projects where lead-based paint or paint with lead must be removed or controlled (including paint film stabilization) or lead-based paint hazards abated as defined by Public Law 102-550 Title X - Residential Lead-Based Paint Hazard Reduction Act of 1992. Use reference (i) for medical surveillance of all lead exposed workers including contract employees.

b. Contract administrators must ensure that each contract, for work performed by an independent contractor in the United States or overseas which may involve the release of lead dust, must incorporate the appropriate references and clauses to ensure that:

- (1) The contractor is aware of the potential hazard to their employees and Navy personnel.
- (2) The contractor complies with reference (a) sections 1910.134, 1910.1025, 1915.1025, 1026.62 and reference (b) and (m) to protect their employees, as well as Navy personnel.
- (3) The contractor must control lead dust outside of the work boundary to less than 30 g/m³ at all times and must perform sufficient monitoring to confirm that this level of control is

maintained. In addition, the controlled work area(s) must meet these criteria prior to release for unrestricted access. Contractors must provide copies of their monitoring results to the cognizant industrial hygienist via the contract administrator.

B2109. Responsibilities.

a. Chief, Bureau of Medicine and Surgery must:

(1) Centrally manage the lead medical surveillance program ashore and afloat as well as maintain an electronic records database to allow for record access and data analysis.

(2) Provide professional IH technical support and training assistance to commands, units and activities for the purpose of evaluating the potential for lead exposure.

b. Commanders of echelon 2 and other headquarters commands must:

(1) Provide advice and technical assistance to define appropriate engineering and work practice controls and to identify acceptable lead free substitute materials.

(2) Ensure program support by budgeting the resources required to meet the regulatory standards for the control of lead as prescribed by this chapter.

(3) Determine the applicability of reference (a) sections 1915.1025 and 1926.62 to any operations within their respective major commands and provide policy and guidance to affected commands, units and activities. This will require procedures to ensure pre-placement medical screening and training are provided to workers based on occupational “task based triggers” specified in reference (a) section 1915.1025 and reference (b).

c. COMNAVFACSYSCOM must:

(1) Provide advice and technical assistance concerning lead paint in Navy buildings, particularly housing, childcare facilities and hospitals.

(2) Ensure that contracting officers and representatives receive the appropriate level of training to adequately plan, design, oversee and review lead construction work.

d. Commanders, Commanding Officers and Officers in charge of commands, units and activities must:

(1) Apply control measures and monitoring procedures prescribed in this chapter to processes using lead or lead containing materials.

(2) Budget resources in order to meet these lead control requirements.